

Hermite 4-Point Cubic Interpolation & Varispeed Resampling

DSP Mathematics of Wavetable, Sampler, and Delay Line Objects

Time Dilation DAW (v4.0) Engineering Group

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Abstract

This paper formulates the mathematical models of sub-sample fractional position interpolation used in **Time Dilation DAW**'s `[osc~]`, `[sampler~]`, `[tabosc4~]`, and `[delay~]` nodes, detailing 4-Point Hermite 3rd-Order Cubic Splines, Linear Interpolation, and Nearest-Neighbor Zero-Order Hold.

1 Fractional Table Read Head Position

When playing audio buffers or wavetables at non-integer pitch ratios or dilated time speeds γ , the fractional read head position $p(n)$ at sample n is:

$$p(n) = p(n-1) + \text{step}, \quad \text{where } \text{step} = \frac{|\gamma(n)| \cdot f \cdot N_{\text{table}}}{f_s} \quad (1)$$

Decomposing $p(n)$ into integer index $i = \lfloor p(n) \rfloor$ and fractional offset $\alpha = p(n) - i \in [0, 1)$.

2 Hermite 4-Point 3rd-Order Cubic Spline Interpolation

To eliminate high-frequency aliasing and inter-sample quantization noise during varispeed pitch shifting, we evaluate four adjacent table samples (y_0, y_1, y_2, y_3) at indices $(i-1, i, i+1, i+2)$:

$$y(\alpha) = c_3\alpha^3 + c_2\alpha^2 + c_1\alpha + c_0 \quad (2)$$

where the polynomial coefficients are computed from the 4-point stencil as:

$$c_0 = y_1 \quad (3)$$

$$c_1 = \frac{1}{2}(y_2 - y_0) \quad (4)$$

$$c_2 = y_0 - \frac{5}{2}y_1 + 2y_2 - \frac{1}{2}y_3 \quad (5)$$

$$c_3 = \frac{1}{2}(y_3 - y_0) + \frac{3}{2}(y_1 - y_2) \quad (6)$$

2.1 Matrix Representation

The Hermite interpolation polynomial can be expressed in compact matrix vector form:

$$y(\alpha) = \begin{bmatrix} 1 & \alpha & \alpha^2 & \alpha^3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 1 & -\frac{5}{2} & 2 & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} \quad (7)$$

3 Linear Interpolation & Nearest Neighbor

3.1 Linear Interpolation (2-Point)

For computationally lightweight playback:

$$y(\alpha) = (1 - \alpha)y_1 + \alpha y_2 \quad (8)$$

3.2 Nearest Neighbor (Zero-Order Hold)

For intentionally quantized glitch sound design:

$$y(\alpha) = y_{\lfloor \alpha + 0.5 \rfloor + i} \quad (9)$$

4 Spectral Error & Aliasing Suppression

Hermite 4-point cubic interpolation provides $O(\Delta x^3)$ local truncation error accuracy, attenuating high-frequency spectral imaging foldback by more than 60 dB across the audible range compared to linear interpolation.